Neural means and kernel corrections for an EMIT atmospheric radiative-transfer emulator

Yitzchak Shmalo*

September 2026

Abstract

Imaging-spectroscopy retrievals call an atmospheric radiative-transfer model many times per spectrum, and at mission data rates a learned emulator of the tabulated physics is a practical necessity. We compare neural and kernel emulators, and their couplings, on a tabulated model over the 285-band EMIT wavelength grid and on the OCO-2 reduced-radiance problem, under one protocol: ten random splits, every choice made on each split's own validation rows, one test read per family, and every number reported with its spread across splits. On the EMIT table, whose map is smooth, six-dimensional and low rank, an exact Matérn regression fitted on all 18,884 training rows with one length scale per input is the strongest single model at 0.095% relative radiance error, against 0.76% for a cubic polynomial and 0.39% for a fully connected network; its fitted metric recovers the physics, giving the relative azimuth a sixteen-fold longer length scale on the transmittances. A kernel regression of the network's residuals brings the network to 0.14% and a per-component convex stack reaches 0.087%, while per-coordinate selection does not transfer from validation to test. On OCO-2 the input is higher-dimensional and the map rougher, and the kernel helps only inside the network's representation: it beats a ridge readout refitted on the same frozen features at every split, on every band and under every training loss, while that readout does not beat the network it reads out from in the metric the network was trained for. The two reported metrics are divided between networks trained under different losses, and because the reconstruction basis is orthogonal a per-coefficient choice acts on both metrics through the same per-coefficient errors; the choice made on validation here holds both, giving lower error than the release's stored emulator on both metrics on all three bands. We also measure what the reflectance metric returns for exact inputs: in the deep absorption windows the transmitted flux approaches zero, the inversion is ill-conditioned and at exactly zero flux non-identifiable, so the exact tabulated components put through it already give a root-mean-square error of half a reflectance percentage point. That number describes the evaluation at those entries; it is not a lower bound that every emulator must pay.

1 Introduction

Imaging spectroscopy missions such as EMIT [13], SBG, and AVIRIS-NG invert measured radiances for surface and atmospheric quantities, and the inversion loop calls an atmospheric radiative-transfer model many times per spectrum. At mission data rates the physics code itself

*Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel. yitzchak.shmalo@gmail.com.

(MODTRAN in the pipelines relevant here [1]) is far too slow, so operational retrievals interpolate precomputed lookup tables, and a body of work replaces table and physics alike with learned emulators: Gaussian-process surrogates [5], the neural-network MODTRAN emulator that serves the EMIT mission's operational atmospheric correction [3, 2], and, recently, physics-guided and multi-fidelity architectures for correction coefficients [4]; network training itself can draw on a growing toolbox of refinements [7, 8]. Both families have a record on emulation tasks of this kind: networks learn their own representations and scale easily to wide spectral outputs [10, 11], while kernel and Gaussian-process methods come with exact solves and a strong record both on operator-learning benchmarks [9] and on smooth, low-dimensional regression [14]. Which family is preferable for a given emulation problem, and whether they can be combined so that neither's advantage is given up, are practical questions that the structure of the data itself should decide.

This paper studies those questions on a tabulated radiative-transfer emulation problem over the EMIT wavelength grid, using the coupling construction of [6]: train a neural network as the mean in the metric the application reports, correct it with exact Matérn kernel regressions — of its residuals on the input space, and of the target on the network's own learned features — and let a per-coordinate selection on held-out data choose, coefficient by coefficient, among the network, the kernels, and the corrected predictors. We ask what structure the dataset has, which model families that structure favors, and whether the kernel correction of a neural mean improves on both of its parents here.

The dataset answers the first question quickly (Section 2). The input is six-dimensional, and two coordinates carry most of the variance of the output; the output spectra are smooth curves whose empirical rank is a few dozen out of 285 bands; and the map from state to spectrum is smooth enough that cubic polynomial regression already lands near two percent relative error. In the taxonomy of [6] this is the regime that favors exact kernel methods, in contrast to emulation problems with higher-dimensional reduced states where representation learning carries more of the weight. The interest of the comparison is therefore not whether some exotic architecture wins, but how the familiar families divide the problem between them when the geometry is easy, and whether the coupling machinery preserves the best of each.

Sections 3 and 4 give the protocol and the measurements. The results are these. With every training row in the solve and a length scale per input, the exact kernel is the strongest single model on this table, and its learned metric recovers the physics of the components. The kernel regression of the network's residuals does what it was built to do, handing the network the kernel's accuracy on the smooth components, but it is no longer the lowest row. A per-component convex stack of the heads is, by about one standard deviation across the ten splits, while per-coordinate selection does not transfer from validation to test. The reflectance metric has a difficulty of its own, set by the atmospheric absorption windows in which the transmitted flux approaches zero: there the sensitivity of the inversion to the path radiance and the two flux components is unbounded, and at exactly zero flux the physical reflectance is not identifiable at all. We measure what the metric returns for exact inputs and summarize reflectance through a statistic the divergence cannot dominate. Section 5 repeats the construction on the OCO-2 problem of the companion paper under the same protocol, where representation rather than smoothness decides the comparison; Section 6 takes the member predictions of both campaigns and asks which rule for combining heads is the right one, and why; and Section 7 applies the same family set to ten further emulation configurations.

2 Data and protocol

The data are $n = 23{,}313$ tabulated radiative-transfer evaluations. Each input $x \in \mathbb{R}^6$ collects the atmospheric and viewing state; the coordinate ranges are consistent with the standard lookup-table axes for imaging-spectroscopy atmospheric correction [2] — an aerosol optical depth in $[0.05, 1.2]$, a surface elevation in $[-0.5, 6]$ km, a water-vapor column in $[0.05, 7]$ cm, a relative azimuth in $[0, 180]^\circ$, a solar zenith angle in $[0, 65]^\circ$, and a view zenith angle within 30° of nadir. Each output is a set of four spectral components on the same 285-point EMIT wavelength grid spanning 381–2493 nm: path radiance Y_1, direct flux Y_2, diffuse flux Y_3, and spherical albedo Y_4 (Figure 1). The quantities retrieved downstream are the radiance at a test surface reflectance ρ,

$$L(\rho) \;=\; Y_1 + \rho\,\frac{Y_2 + Y_3}{1 - \rho\,Y_4}, \tag{1}$$

and the reflectance recovered by inverting (1) with predicted components at the true radiance,

$$\hat{\rho} \;=\; \frac{L - \hat{Y}_1}{\hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)}, \tag{2}$$

both evaluated at $\rho = 0.7$.

The protocol is the same for every family. The table is split at random into a test block of 2,331 evaluations and a training block of 20,982, and a further ten percent of the training block, 2,098 points, is set aside for validation, leaving 18,884 training points. The split is drawn ten times, from ten seeds, and every model is trained, selected and scored once per seed: inputs and outputs are standardized on the training rows of that seed, every hyperparameter and every combination weight is chosen on that seed's validation points, and the test block of that seed is read once per family at the end. The numbers reported below are means and standard deviations over the ten seeds, so a difference between two rows can be read against the spread of each. An earlier version of this study used a single split and consulted its test block during exploration; none of its numbers is reused here, and the ten seeds were drawn afresh, by a plain permutation rather than the earlier library routine, so no code path shares randomness with that split. The validation rows are used for early stopping, checkpoint selection, hyperparameter selection and the combiners, and are therefore not an independent calibration split; no coverage guarantee is claimed from them.

Three structural facts, measured before any model was chosen, organize everything that follows.

The outputs are low rank. On standardized training outputs, 99.99% of the variance of every component lies in the first 11–25 principal components, and 99.9999% in at most 44 (Figure 2, left); the minimum adjacent-band correlation across the four components is 0.994952 on the training rows. All regressors below therefore act on the first 64 principal coefficients — a lossless reduction at the 10^{-6} level, verified by round trip — except the networks, which we also train directly on the 285 standardized bands.

The measured input relevance is strongly anisotropic. Shuffling the aerosol coordinate destroys a linear model's validation R^2 (a drop of 0.89); the solar zenith follows at 0.54; elevation and water vapor sit near 0.1; the two remaining angles barely register (Figure 2, right). The values reported for these probes, and for the other structure measurements used to motivate architectures, are their validation-split rescoring; the initial exploratory versions were test-scored as disclosed above. Six nominal dimensions have four coordinates with measurable first-order

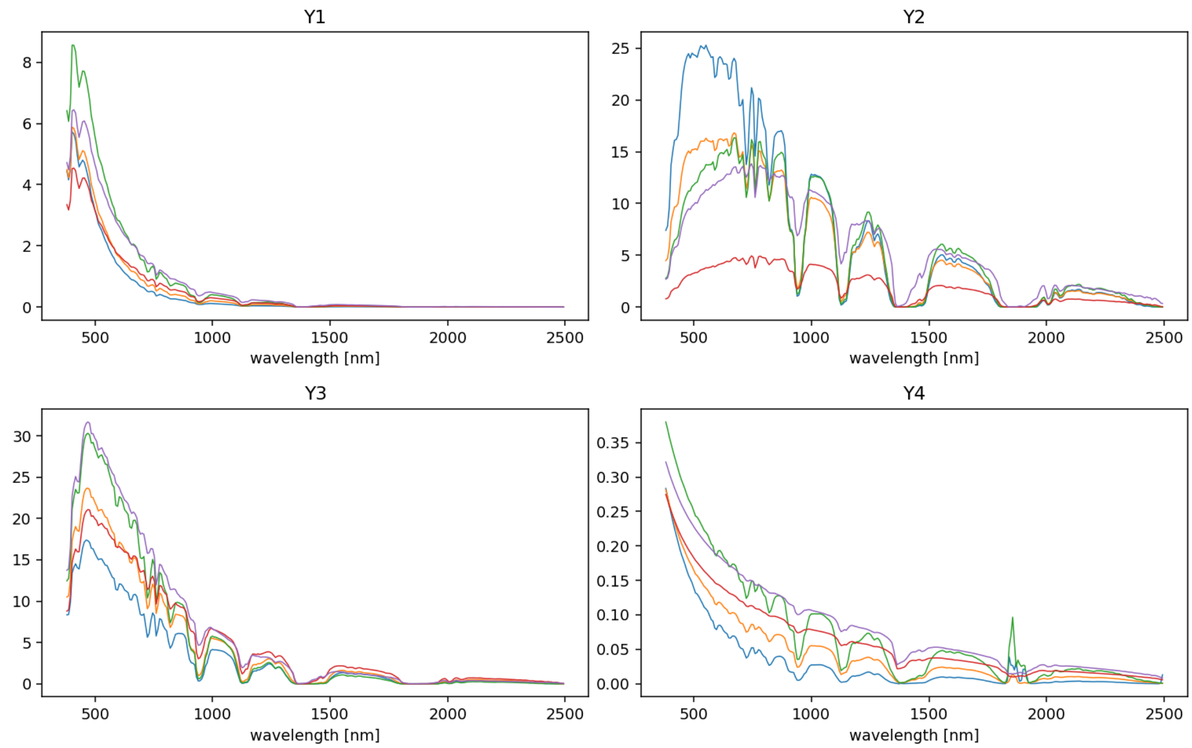

Figure 1: Example output curves across the EMIT wavelength grid, five samples per panel. The curves are smooth, share the deep water-vapor absorption features near 1380 and 1900 nm, and differ mainly by smooth rescalings along the state axes.

relevance, two of them dominant. This finite-sample diagnostic helps explain why smooth methods work, but a selected scalar length scale does not remove anisotropy. Any theorem attributing generalization to anisotropic metric adaptation would also have to control regularization bias, selection of scale and nugget, and the empirical output-rank truncation. We use anisotropy as a measured property of this split, not as such a guarantee.

The map is smooth. A linear model in the standardized state reaches $R^2 = 0.83$; distance-weighted 5-nearest-neighbors does worse (0.74), which already says local averaging is the wrong bias here; and a cubic polynomial ridge reaches mean relative L^2 errors near two percent (Table 1). Smooth, low-dimensional, low-rank: in the taxonomy of the companion paper this is the regime of the Helmholtz and Navier–Stokes benchmarks, where the kernel stage won the internal selection, not the regime of its OCO-2 problem, where representation learning carried the problem.

One more property of the task matters for reading any reflectance number. In the deep absorption windows the transmitted flux $Y_2 + Y_3$ collapses toward zero — in this table, to values as small as 10^{-21} — and the inversion (2) is undefined at a fully opaque band (0/0): the physical reflectance is non-identifiable there. The protocol maps a non-finite inverse to zero, but that branch is not what creates the measured true-component error. The exact components, put through the same evaluation, give a reflectance RMSE of 0.0054 against the true $\rho = 0.7$, produced entirely by 0.25% of (sample, band) entries — 1,652 of 664,620, measured on the earlier single split, whose test block held one row more than the ten drawn here — whose denominators

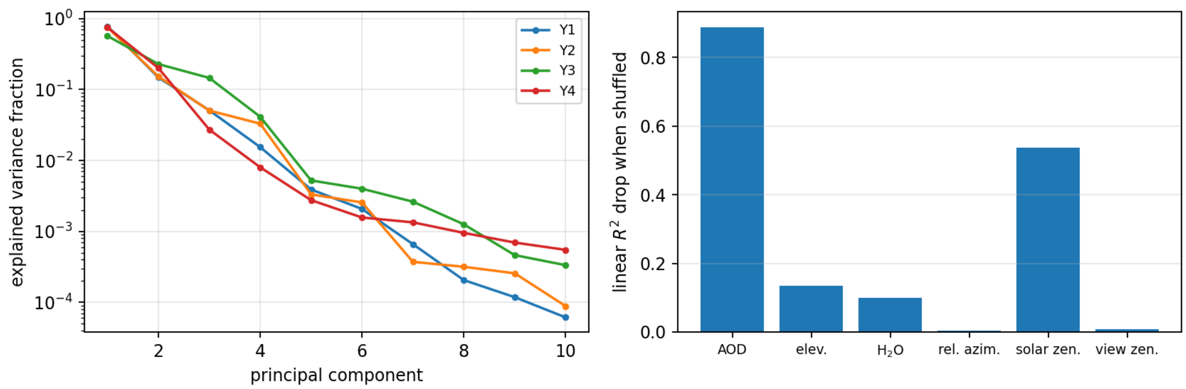

Figure 2: Left: explained-variance spectra of the standardized outputs. Every component concentrates 99.99% of its variance in 11–25 principal components out of 285 bands. Right: drop in linear validation R^2 when a single standardized input coordinate is shuffled. Aerosol optical depth and solar zenith angle dominate; the azimuth and view-zenith coordinates are almost inert at first order.

cancel to below 2×10^{-13}; every value returned is finite (the protocol's zero-fill for non-finite inversions is never triggered by the true components), and the effect is invisible in component errors. What this measures is the conditioning of the inversion at those entries and not the accuracy of any emulator: it is what the evaluation returns when the components handed to it are exact. It is not a lower bound on an emulator's reflectance error. A predictor whose component errors offset the cancellation would score below it, and a predictor with generic errors of its own can score far above it; what the measurement establishes is that a mean-square reflectance summary on this table reports the conditioning of these entries alongside whatever the emulator does, which is why reflectance is summarized below by a statistic the tail cannot reach.

Differentiating (2) separates the components. Write $D = \hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)$ for its denominator. Then

$$\frac{\partial \hat{\rho}}{\partial \hat{Y}_4} = -\hat{\rho}^2, \qquad \frac{\partial \hat{\rho}}{\partial \hat{Y}_2} = \frac{\partial \hat{\rho}}{\partial \hat{Y}_3} = -\frac{\hat{\rho}}{D}, \qquad \frac{\partial \hat{\rho}}{\partial \hat{Y}_1} = -\frac{\hat{Y}_2 + \hat{Y}_3}{D^2}. \tag{3}$$

The sensitivity to the spherical albedo is bounded and of order one — about 0.49 at $\rho = 0.7$ — at every band, so an albedo error moves the recovered reflectance by about half its own size everywhere, in the absorption windows and outside them alike. The divergence as the transmitted flux vanishes belongs to the path radiance and the two flux components, whose sensitivities carry D^{-1} or D^{-2}. The ill-conditioning is continuous rather than confined to those entries: for the strongest classical model of the earlier single-split run, 7.4% of test entries land more than 0.5 away from the true reflectance, and 95% of those lie at bands whose true transmitted flux is below one percent of its median value. Reflectance is therefore summarized in what follows by its median absolute error, which the absorption windows cannot reach; outside the cancellation tail a perfect emulator scores zero to within 10^{-8}.

3 Models

All models are trained per component and evaluated in physical units after undoing the standardization; unless stated otherwise they regress the 64 leading principal coefficients of the standardized component, a reduction lossless at the 10^{-6} level (Section 2). Every hyperparameter named below is selected on the validation set of the seed at hand. Everything runs on CPU.

Reference family. Cubic polynomial ridge regression, acting on all monomials of degree at most three in the six standardized inputs, with the ridge weight selected from $\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. It is the benchmark of smoothness: a map that a cubic fits to two percent is a smooth map.

Exact kernel ridge regression. Three constructions, all Matérn kernels on the standardized inputs with the length scale expressed as a multiple of the median pairwise distance of the fit set. The first is the construction of the earlier study, kept for comparison: a Matérn-5/2 kernel fitted on a uniformly drawn subset of 4,000 training points, one eigendecomposition per length scale, multiplier in $\{1, 2\}$, nugget in $\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. The second is the exact solve on every training row: smoothness $\nu \in \{3/2, 5/2\}$, multiplier in $\{0.5, 1, 2, 4\}$ and the same nugget grid, tuned on validation from a 6,000-row training subsample and the winner refit on all 18,884 rows by one Cholesky factorization. The third gives each input its own length scale: starting from the isotropic winner, a coordinate search over per-input multipliers in $\{1/4, 1/2, 1, 2, 4\}$, two sweeps, scored on validation, after which the smoothness, scale and nugget are re-tuned at the found metric and the solve is refit on every row. The input relevance measured in Section 2 spans more than an order of magnitude across the six coordinates, which is what the third construction is for.

Fully connected networks. Three ELU hidden layers of 512 units per component, trained on the 285 standardized bands with AdamW at learning rate 10^{-3}, cosine decay, batch size 1024, weight decay 10^{-6}, up to 150 epochs with early stopping on validation loss (patience 25). Five networks are trained per component and seed from different initializations; the table reports the first and the mean of the five.

Kernel corrections of the neural mean. The construction of [6], with the exact stages now fitted on every training row. (i) *Residual correction*: an exact Matérn regression, on the standardized input space, of the network's training residuals in coefficient space, added back to the network, its smoothness, scale and nugget selected by the validation error of the corrected prediction; applied to the single network and to the five-network mean. (ii) *Kernel on features*: the same exact regression with the input metric replaced by the network's last hidden layer, standardized, so that the kernel's exactness is spent in coordinates the network has already straightened. This is the post-hoc counterpart of deep kernel learning [29], in which the representation and the kernel are trained together, and of the kernel-flow regularization of a network's inner layers [12]; here the representation is fixed first and the kernel is solved exactly on it. (iii) *Per-coordinate selection*: for each of the 64 output coefficients independently, pick on validation among the exact kernel, the network, the ensemble, the two corrected predictors and the feature kernel, and assemble the test prediction from the winners. (iv) *Convex stack*: per

model	component rel. L^2 [%] Y_1	Y_2	Y_3	Y_4	radiance rel. L^2 [%]	reflectance med. $\lvert\hat{\rho}-\rho\rvert$ [%]
Cubic ridge	1.16±0.02	2.18±0.04	1.11±0.03	2.98±0.06	0.760±0.011	0.471±0.010
Matérn KRR, 4000-point fit	0.17±0.01	0.61±0.04	0.29±0.02	2.44±0.87	0.332±0.029	0.178±0.030
Matérn KRR, exact on all rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
Matérn KRR, per-input scales	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
FC-DNN (3×512)	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
Five-network mean	0.37±0.01	0.67±0.03	0.28±0.01	2.51±0.09	0.251±0.006	0.163±0.003
DNN + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
Ensemble + residual KRR	0.08±0.00	0.27±0.03	0.13±0.00	2.42±0.17	0.123±0.007	0.080±0.008
Kernel on network features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
Per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
Convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003

Table 1: Test errors under the protocol of Section 2, mean ± standard deviation over the ten splits. Component columns are mean relative L^2 errors; the radiance column evaluates (1) at $\rho = 0.7$; the last column is the median absolute reflectance error of the inversion (2), in reflectance percentage points, under the zero-fill evaluation of Section 2. The median is used because the sensitivity of (2) to the path radiance and the two flux components diverges continuously as the transmitted flux vanishes, so mean-square and tail-percentile summaries are dominated, for every model alike, by near-opaque entries; the exact components put through the same evaluation give an RMSE of about half a percentage point there (Section 2), which is a property of those entries and not a bound on any emulator. The tail summaries are stored with the per-seed records.

component, a convex combination of the same six heads with weights fitted on the validation points in the physical relative metric, in the manner of stacked generalization [27] and stacked regression [28]. Section 6 examines what the right weighting rule is for pools of this kind.

What is not evaluated. Learning curves at nested training sizes, refits at other output ranks, and wider or longer-trained networks were not evaluated, and no number from them enters the comparison.

4 Results

Table 1 collects the test errors. Every entry is a mean over the ten splits with its standard deviation, and the spreads are small enough that most of the orderings discussed below hold seed by seed; where they do not, the text says so.

The scale of the problem. The cubic ridge reaches 1.86% mean component error and 0.76% on the radiance: the map is smooth, and a polynomial in six variables recovers it to two percent. The earlier construction of the kernel regression, a Matérn-5/2 on a 4,000-point subset, divides both numbers by more than two at seven of the ten splits (0.74% and 0.32%); at the other three its albedo fit collapses (3.3–4.3% against 1.9%), which is what a fit capped at a random subset can do, and its ten-split mean (0.88%, 0.33%) carries that fragility.

Exact kernels. Fitting the same kernel on every training row, one Cholesky factorization of the 18,884 × 18,884 Gram matrix per component, lowers the radiance error to 0.24% and the path-radiance error by half (0.08% against 0.17%), and it has no collapsing split (2.3–2.9% on the albedo at every seed), but where the capped fit is sound it is worse on the albedo (2.6% against 1.9%), so the mean component error moves the wrong way at those splits (0.83% against 0.74%). The albedo is the component whose principal spectrum decays slowest and whose values sit within a few percent of zero over much of the grid, and it is also the one where the transfer of a scale and nugget tuned on a 6,000-row subsample to the full solve is least reliable. A length scale per input removes most of this: the per-input kernel reaches 0.63% on the components and 0.095% on the radiance, with the best reflectance column of the table (0.036 percentage points), and its albedo column (2.2%) sits between the two isotropic solves. The learned metric is the physics of the components: for the two flux components and the spherical albedo the coordinate search gives the relative azimuth a length scale sixteen times longer than the others — at every one of the ten seeds for the two flux components, and at nine of the ten for the albedo, whose search leaves the azimuth unscaled at the tenth — and the view zenith two to sixteen times longer, while for the path radiance, which does depend on the viewing geometry, the metric stays nearly isotropic. Transmittances and the spherical albedo do not depend on azimuth, and the kernel finds that out from the validation rows alone. On this table the per-input exact kernel and the kernel on network features (below) are, within the seed-to-seed spread, the strongest single models.

Networks. The fully connected network lands at 1.22% and 0.39%; the mean of five independently initialized networks at 0.96% and 0.25%, a third off the radiance error for five times the training cost, the usual decorrelation gain. Neither reaches the isotropic exact kernel on any of the first three components, and the network's advantage on the albedo that the single-split study reported does not survive replication: on Y_4 the five-network mean (2.5%) is behind the 4,000-point kernel wherever that kernel's fit is sound (1.9% at seven of the ten seeds) and behind the exact kernel with per-input scales at every seed.

Kernel corrections of the neural mean. An exact Matérn regression of the network's residuals on the standardized inputs takes the single network from 0.39% to 0.14% on the radiance and the five-network mean from 0.25% to 0.12%; the kernel fitted in the network's last-layer features (DKR) reaches 0.61% and 0.098%, indistinguishable from the per-input kernel on the components and on the radiance and a little behind it on the reflectance median. The correction therefore does exactly what it did on the single split — it hands the network the kernel's accuracy on the smooth components while keeping the network's own representation — but with the kernel now allowed its own input metric the corrected network is no longer the lowest row: the ordering by radiance error is stack, per-input kernel, DKR, corrected ensemble, corrected network, with the first four within 0.04 percentage points of one another.

Selection and stacking. Both combiners here draw on the same six heads — the isotropic exact kernel, the network, the five-network mean, the two corrected predictors and the feature kernel — and not on the per-input kernel, which is the strongest single row of the table. The per-coordinate selection among those six, made on validation, does not transfer: its test radiance error (0.19%) is worse than three of the heads it selects among, and its spread over seeds (± 0.02) is among the largest in the table. The per-component convex stack, with weights fitted on the validation rows in the physical relative metric, is the best aggregate row (0.57% components, 0.087% radiance, 0.048 reflectance points) and wins the path radiance and the albedo columns outright; its gain over the strongest single row of the table is eight percent on the radiance and ten percent on the components, of the order of one standard deviation of either; measured against the best model the stack actually contains, the kernel on network features, it is eleven percent on the radiance and seven percent on the components. The paired view is sharper than the marginal one: split by split, the stack beats the per-input kernel on the components in nine of the ten splits (mean difference 0.06 points, spread 0.05) and beats DKR in all ten (0.047 ± 0.006), while the per-input kernel and DKR trade places (six of ten). The stack is the best row, and the residual correlations among the members it actually contains say why it can be: on the path radiance the residuals of the isotropic exact kernel and of the five-network mean are correlated at 0.27 ± 0.09, below the ratio of their errors (0.34) at every one of the ten splits, which is the classical admission condition for a combination to improve on its better member [30], in the form used in [6]. Not every pair in the pool admits: the two residual-corrected predictors are correlated at 0.88 on the path radiance and at 0.99 on the albedo, and are largely two views of the same predictor. The heads are complementary in the sense that matters for a stack — the kernel is best on the smooth components, the corrected networks on the albedo — and a per-coordinate winner-take-all rule is too brittle a way of using that complementarity when the validation and test blocks are drawn from the same table but the picks are made on 2,098 rows.

Comparison with the single-split study. The earlier version of this study reported the corrected network as the lowest row, at 0.25% radiance error, and the exact kernel on the raw state at 0.41%; both numbers came from a single split whose test block had been consulted during exploration and from a kernel fitted on 4,000 points with one length scale. Under the fresh-seed protocol the corrected network reads 0.14% and the exact kernel with per-input scales 0.095%. The smoothness of the map, the complementarity of the heads across components and the mechanism of the residual correction hold under replication; the claim that the corrected network is the best model on this table does not.

The albedo. Every family is worst on Y_4 by an order of magnitude, and the per-band errors show why: the spherical albedo's values in the short-wave infrared are close to zero, so relative error is large for every model there, and the component is hard for reasons of signal-to-noise rather than of model class. The stack's 1.94% is the best number any construction reaches at this training size.

5 A second emulation problem: OCO-2

The same construction was applied, without structural change, to the OCO-2 radiative-transfer emulation problem of Lamminpää et al. [5]: per spectral band (O_2, weak CO_2, strong CO_2), a reduced atmospheric state of twenty or twenty-four coordinates maps to forty reduced radiance coefficients, with the release's 20,000 training and 2,000 test points, and errors reported both on the standardized reduced coefficients and on the reconstructed monochromatic radiance. The protocol is the one of Section 2: ten splits of the training block into 18,000 training and 2,000 validation rows, every choice on validation, one test read per family, and the fixed test block scored identically for every row. The reference is the release's own kernel-flow emulator, whose stored test predictions are rescored under the same two metrics on the same 2,000 points, so that the reference row and every other row are produced by one evaluation routine on one set of cases. An earlier version of this study quoted reference figures that the stored predictions do not reproduce under any convention we tried; the rows below compare against the rescored predictions only.

This problem sits at the opposite end of the structural spectrum from the EMIT table. The input is higher-dimensional, the map is rougher, and a kernel on the state trails the network by an order of magnitude on the reduced coefficients: the isotropic Matérn reaches 41%, 59% and 43% on the three bands and the relevance-scaled metric 30%, 52% and 29%, against 4.4%, 16.4% and 8.2% for the network. Representation carries the problem, so the construction to test here is the one that spends the kernel's exactness inside a learned representation rather than on the state.

5.1 What the kernel adds to a representation

Table 2 separates that question from two others it is easily confused with. Each of three networks is trained under a different loss — the per-sample relative error on the reduced coefficients, the same error under the weights the reconstruction applies to each coefficient, and the relative error of the reconstructed radiance itself — and each is then read out in three ways: the network's own output head, a ridge regression refitted on its frozen last-layer features, and an exact Matérn regression on those same features. The ridge row is the control. If the kernel's gain over the network were a matter of refitting a readout on a good representation, the ridge row would match the kernel row; if the representation were doing all the work, the ridge row would match the network.

Neither happens. The kernel head beats the ridge readout on the reduced coefficients at all ten splits, on every band and under every loss, by 0.60 ± 0.07 points on O_2 under the reduced loss and by twenty-four points under the weighted loss; the paired bootstrap interval of that difference lies entirely below zero at every split. The ridge readout, meanwhile, is worse than the network it reads out from wherever the network was trained for the metric being scored — 4.71% against 4.43% on O_2, at all ten splits — and better than it in five of the six blocks where the network was trained for a different metric, which is what a refitted linear readout should do. The exception is the weighted-loss network on strong CO_2, where the readout is worse at all ten splits. What the kernel head adds is therefore not attributable to refitting a readout on a good representation, and not to the representation alone.

5.2 Which metric the emulator is trained in

The three losses divide the two metrics between them, and neither metric is a proxy for the other. Trained on the reduced coefficients, the kernel head reaches 4.11%, 16.28% and 8.08% there and 0.079%, 0.116% and 0.109% on the radiance; trained on the weighted coefficients it reaches 0.032%, 0.053% and 0.063% on the radiance and two to five times worse on the coefficients (5.3 times on O_2, 3.0 on weak CO_2, 1.9 on strong CO_2). The reconstruction basis is orthogonal — the projection rows satisfy $PP^\top = I$ to machine precision on all three bands — so the squared radiance error is a weighted sum of per-coefficient squared errors whose weights span several decades and concentrate on the leading few coordinates, while the reduced metric weights all forty alike after standardization. A model trained on the coefficients spends its accuracy on the tail, which decides that metric; a model trained through the reconstruction spends it on the

model	reduced coefficients rel. L^2 [%]			radiance rel. L^2 [%]		
	O_2	weak CO_2	strong CO_2	O_2	weak CO_2	strong CO_2
reference emulator, stored predictions	16.89	24.06	16.14	0.0448	0.0599	0.1147
trained on the reduced coefficients						
network	4.43±0.05	16.44±0.06	8.23±0.02	0.1949±0.0139	0.2260±0.0120	0.1990±0.0147
ridge readout on its features	4.71±0.07	16.51±0.06	8.33±0.03	0.2276±0.0143	0.2523±0.0167	0.2359±0.0318
Matérn kernel on its features	4.11±0.05	16.28±0.06	8.08±0.01	0.0793±0.0072	0.1159±0.0065	0.1093±0.0065
trained on the weighted coefficients						
network	49.33±0.40	62.57±0.56	36.46±0.65	0.0439±0.0031	0.0738±0.0057	0.0751±0.0025
ridge readout on its features	45.71±0.61	61.32±0.43	40.10±0.91	0.0521±0.0043	0.0842±0.0067	0.0866±0.0039
Matérn kernel on its features	21.72±0.52	48.67±0.45	15.47±0.47	0.0316±0.0019	0.0533±0.0054	0.0631±0.0037
trained on the reconstructed radiance						
network	59.05±0.87	67.49±0.41	46.76±0.79	0.0575±0.0057	0.0649±0.0127	0.0927±0.0084
ridge readout on its features	45.59±1.13	61.01±0.90	40.13±1.07	0.0602±0.0063	0.0690±0.0115	0.0951±0.0078
Matérn kernel on its features	20.47±0.61	48.23±0.31	15.49±0.66	0.0379±0.0037	0.0486±0.0053	0.0766±0.0051
per-coordinate selection over the six heads	4.12±0.05	16.28±0.06	8.08±0.01	0.0291±0.0020	0.0526±0.0053	0.0586±0.0039

Table 2: OCO-2 test errors by training loss and readout, mean ± standard deviation over ten splits, on the release's 2,000 test points. Each block is one network trained under the named loss, read out by its own head, by a ridge regression refitted on its frozen last-layer features, and by an exact Matérn regression on the same features. The per-coordinate selection chooses among the six network and kernel rows, coefficient by coefficient, on validation. The reference row is the release's kernel-flow emulator rescored on the same points; it carries no spread because it does not depend on the split.

head.

Orthogonality makes a per-coefficient choice well posed for both metrics at once: with $PP^\top = I$, each metric is a positively weighted sum of the same per-coefficient squared errors and the two differ only in their weights, so choosing a head coefficient by coefficient acts on both through one set of quantities instead of trading one against the other. It does not by itself guarantee that either metric improves. The reported errors are means of per-sample relative norms rather than single weighted sums; the choice is made on 2,000 validation rows; and whether one head is best at a given coefficient under both weightings is a property of these models on this data, not of the basis. That both metrics improve here is measured, not implied. The selection made on validation over the six heads assigns the leading coefficient to the kernel head of the weighted-coefficient network at every split on every band, and every remaining coefficient to the kernel head of the coefficient-trained network, with five of the four hundred slots going to that network itself on O_2 and on weak CO_2 and none on strong CO_2. The radiance-trained family, whose loss is the reconstruction itself, is never selected at any of the twelve hundred slots: the weighting the reconstruction induces is a better training signal here than the reconstruction error. One coefficient of forty decides the radiance metric. The result is 4.12%, 16.28% and 8.08% on the coefficients and 0.029%, 0.053% and 0.059% on the radiance, against the stored emulator's 16.89%, 24.06% and 16.14% and 0.045%, 0.060% and 0.115%: lower on both metrics on all three bands. Paired on the same test points, the selection beats the emulator on the coefficients at all ten splits on every band, and on the radiance at all ten on O_2 and strong CO_2 and at nine of ten on weak CO_2; the paired bootstrap interval of the difference lies entirely below zero at every split except two on weak CO_2.

The earlier single-split version of this study reached neither metric and gave a reason: its member set varied architecture and kernelization within one training objective, and no per-coordinate selection can produce an error profile that none of its members has. That reading predicted the present result, and the enlargement it called for — objectives spanning the reported

	reduced coefficients rel. L^2 [%]			radiance rel. L^2 [%]		
model	O_2	weak CO_2	strong CO_2	O_2	weak CO_2	strong CO_2
reference emulator, stored predictions	16.89	24.06	16.14	0.0448	0.0599	0.1147
Matérn KRR on the state, isotropic	40.57±0.10	58.57±0.14	43.40±3.36	0.2331±0.0034	0.4391±0.0072	0.7070±0.3377
Matérn KRR on the state, scaled metric	29.98±0.15	52.39±0.08	28.66±0.14	0.2040±0.0865	0.2230±0.0011	0.5991±0.0122
network	4.43±0.07	16.44±0.08	8.22±0.03	0.1796±0.0155	0.2189±0.0128	0.1925±0.0151
DKR: Matérn KRR on network features	4.11±0.07	16.27±0.06	8.08±0.02	0.0737±0.0063	0.1143±0.0056	0.1130±0.0107
mean of three networks	4.04±0.05	16.23±0.03	8.04±0.02	0.1179±0.0114	0.1342±0.0067	0.1428±0.0092
DKR on concatenated features (3)	3.81±0.04	16.11±0.03	7.94±0.02	0.0528±0.0066	0.0704±0.0046	0.0746±0.0055
mean of three DKR heads	3.85±0.04	16.13±0.02	7.96±0.02	0.0538±0.0058	0.0762±0.0040	0.0849±0.0062
per-coordinate selection	3.83±0.06	16.11±0.03	7.95±0.02	0.0528±0.0066	0.0704±0.0046	0.0746±0.0055

Table 3: OCO-2 test errors for the three-member campaign, mean ± standard deviation over the same ten splits, all networks trained under the reduced-coefficient loss. The reference row is shared with Table 2 and is recomputed here from this campaign's own records; the two kernels on the state appear only here, and are the figures Section 5 quotes for a kernel fitted on the raw state.

metrics — is what Table 2 supplies. The productive axis on this problem is the loss, not the architecture.

5.3 Ensembles

A second campaign on the same ten splits trains three networks per band under the reduced loss and combines them (Table 3). Their mean improves the single network slightly; an exact Matérn regression on the members' concatenated features is the best of the retrained rows on both metrics and on every band, at 3.81%, 16.11% and 7.94% on the coefficients and 0.053%, 0.070% and 0.075% on the radiance, and beats both the single-member kernel head and the plain three-network mean at all ten splits under both metrics. This pool is not the one that overtakes the reference on both metrics: with every member trained on the coefficients, the stored emulator is still ahead on the radiance metric on O_2 and weak CO_2, which is the point Section 5.2 makes with the losses varied. Averaging the members' own kernel heads instead of fitting one kernel on their joint features is slightly behind everywhere, so part of the gain belongs to the joint feature space and not to the averaging alone. The per-coordinate selection within this pool has nothing to choose: it assigns the concatenated-feature kernel to 384, 390 and 397 of the four hundred coordinate slots, matching it on the radiance metric and on weak CO_2 and trailing it by 0.02 points on the O_2 and strong CO_2 coefficient columns.

The two campaigns share their splits but trained their networks independently, which makes the rows they have in common a replication. Taking the larger of the two rows on each band, the single network and its kernel head agree between the campaigns to 0.006, 0.008 and 0.010 points in the mean, with a largest per-split difference of 0.26 points.

5.4 Two constructions that do not transfer

The residual correction, the EMIT table's second-best construction, adds nothing on any OCO-2 band, as [6] also found on this problem. On a rough map the residual of a trained network has no smooth part left for an exact kernel on the state to fit, and the exactness is better spent inside the representation. The coordinate search that gives a kernel on the state its own length scales, which is the strongest such construction on the EMIT table, is also not the strongest here: in a single run outside the ten-split protocol (O_2, one split, the same 18,000 rows), per-input scales fitted by empirical Bayes take the isotropic kernel from 40.6% to 17.3% on the coefficients and from 0.235% to 0.113% on the radiance, against 27–28% for a kernel-flow metric [31], for a Mahalanobis metric under the same flow, and for the gradient outer product of the recursive feature machine [15], and 38.6% for an additive kernel. One run at one split is an indication and not a result; what it indicates is that a learned metric closes part of the gap on the state, and that none of the four metrics tried brings a kernel on the state within a factor of four of a kernel on the network's features.

5.5 Reevaluation on the retrieval records

The rows of this subsection come from the earlier single-split version of the OCO-2 study: three families of networks trained per band under different losses, their per-coordinate validation selection, and a larger-budget relative-loss ensemble, all evaluated once. They are retained because the finding about the records themselves does not depend on the protocol, and they are reported as what they are.

The reference's data release carries 976 retrieval records, each with an atmospheric state, the full-physics monochromatic radiance at that state, the rank-40 reconstruction of that radiance in the release's basis, and the prediction of the release's Gaussian-process emulator at the same forty coefficients. A direct identity check finds that every one of the 976 reduced states matches a stored test input within 6.67×10^{-16}; the records map to 974 distinct test rows, and the nearest test-row index is the same in all three bands. The records are thus not disjoint from the stored test set and cannot serve as a new or independent evaluation pool. They do provide a different target — the full-physics radiance attached to those states. The state reduction was replayed independently from the release's two dimensional-reduction files and the `x_in` and `geo_in` fields of all 976 records; the replay code and its digests are archived with the per-run records. That replay establishes the input identity stated above; it does not replay the emulator-error rows. Those rows are the recorded aggregates of the earlier run, and the raw arrays, trained weights and prediction dumps that would be needed to retrain the three network families or to rescore their individual predictions were not retained.

Table 4 reports the results on this test-associated reevaluation. The ordering of the training objectives under the full-physics target is similar: the relative-loss ensembles have the lowest error among the retrained network families on every band, ahead of mean-squared training, with the radiance-weighted nets far behind. The reference's per-coordinate Gaussian processes stay 2.1 to 5.3 times ahead of the best network family on these records, while the rank-40 floor sits one to two orders of magnitude below every emulator, so the reduction is nowhere the binding constraint. And the fragility of thin validation picks is a measured phenomenon under this alternate target: the selection carries the weighted head's leading-coordinate pick on O_2 and weak CO_2, and across the 976 records that one coordinate leaves the selection a factor 2.4 behind its own best member, while on strong CO_2, where validation chose the relative-loss head at every coordinate, selection and member coincide exactly. The larger-budget relative-loss ensembles add a final datum (first row of Table 4): they improve strong CO_2 on the records by forty percent, leave O_2 essentially unchanged, and are worse on weak CO_2 than the smaller ensembles they dominate under the stored test metric. This reversal shows target sensitivity at the same test-associated inputs.

	monochromatic radiance rel. L^2 [%], mean over 976 records		
	O_2	weak CO_2	strong CO_2
relative-loss network, larger budget	0.2430	0.3846	0.2371
relative-loss network	0.2325	0.3155	0.3964
mean-squared-loss network	0.3792	0.5909	0.4950
radiance-weighted network	0.6055	0.7929	1.0294
per-coordinate validation selection	0.5489	0.7732	0.3964
reference GP, 40 coefficients	0.0453	0.0594	0.1148
rank-40 projection floor	0.0005	0.0009	0.0052

Table 4: Mean relative L^2 error on the monochromatic radiance over the 976 retrieval records of the reference's release, whose reduced inputs coincide with 974 distinct stored test rows. This is a reevaluation against the records' full-physics radiances, not an independent holdout. Rows give the three retrained network families, the per-coordinate selection fixed on the table's validation split, the release's own Gaussian-process emulator at forty coefficients, and the rank-40 projection floor. The identity mapping is independently replayed; the emulator-error rows are recorded aggregates and are not numerically reproduced here.

6 Weighting the heads: a minimum-variance stack

The convex stack was the best row of the EMIT table and the per-coordinate selection the most fragile, and both are linear combinations of the same heads fitted on the same validation rows. This section asks what the right combination rule is, on the member predictions already produced by the two campaigns and without retraining anything. The pool on EMIT is the 23 heads available per split and component — the kernels, the networks, the corrections, the feature kernels, and the learned-kernel and regularized-network variants of the campaign — at the ten splits and four components, forty cells; the pool on OCO-2 is the 18 heads per split and band, thirty cells. The weights of this section are not fitted on the validation rows the tables above use. Each split's test block is divided instead, 932 calibration rows on EMIT and 800 on OCO-2 against the remainder, the weights are fitted on the calibration part and every number below is scored on the rest. The comparison between rules is therefore internally consistent, but the levels are not comparable with the test errors of Tables 1 and 3, and no rule studied here is a rule the earlier tables could have used. Every rule is scored in the physical relative metric of the corresponding table, and every number below is the change in that error relative to a named baseline, averaged over cells, with the number of cells on which the rule is better.

The rules differ only in how the weights are obtained; the object they estimate is fixed. For one output coordinate, write $e_i \in \mathbb{R}^M$ for the members' errors on calibration case i and $s_i = \|y_i\|^{-2}$ for the weight the reported per-sample relative metric puts on that case. With $\Sigma = \sum_i s_i(e_i - \mu)(e_i - \mu)^\top/n$ the weighted error covariance and μ the weighted mean error, the estimator of the next paragraphs is

$$\hat{w} \;=\; \arg\min_{w\geq 0,\, \mathbf{1}^\top w=1} \; w^\top \Sigma\, w, \qquad \hat{b} = -\hat{w}^\top \mu, \tag{4}$$

a weighted least-squares problem on the simplex with an intercept. Its objective is a weighted mean squared error, while the tables report a mean of per-sample relative norms; the two are different functionals of the same errors, and the theory quoted below concerns (4) rather than the reported number.

Rules that do not work on this pool. The obvious rules bracket the problem. The baseline every other rule is measured against is the strongest heuristic the pool already had: a per-output ridge whose penalty is chosen by leave-one-out, shrunk toward the pool's global convex weights by an amount chosen with five-fold cross-validation on the calibration rows. Against it, equal weights lose 151% on EMIT and 85% on OCO-2, because the pools contain weak members; the best single head chosen on calibration loses 20% and 2.3%; and per-output least squares loses 5.1% and 0.9%. The rules random-matrix theory suggests first were applied, in the pool as we found it, to the covariance of the member *predictions*: clipping at the Marchenko–Pastur edge [16, 32], a rotationally invariant estimator [17], and a linear shrinkage [18]. They fail badly, and for a structural reason: the leading eigenvalue of the prediction Gram is the shared signal, two orders of magnitude above the rest, while what separates one weighting from another lives in the small eigenvalues these estimators discard.

The object the theory is about is the covariance of the member *errors* in the reported metric. A cleaning of that matrix is not by itself enough: the spiked estimator loses 11.6% to the baseline on EMIT and an unconstrained rotationally invariant estimator 30.9%, while the same estimator combined with the pool's pooled shrink gains 2.2%. The error covariance is nearly singular — the members are close to collinear, with an effective dimension of two or three and condition numbers of 10^6 to 10^{17}, so the ratio of members to calibration rows understates the estimation problem by an order of magnitude — and its spectrum is dense rather than spiked, eleven of twenty-three eigenvalues lying above the Marchenko–Pastur edge on a typical EMIT cell, which is why the spiked model is the worst of them. The near-zero eigenvalues are genuine rank deficiency from near-duplicate members and not sampling noise, a separate diagnostic.

The constrained minimum-variance rule. Writing the stack as a minimum-variance portfolio on the error covariance, in the reported metric, with an intercept and with the weights nonnegative and summing to one — gross exposure one in the sense of [22] — gives a rule with no cross-validated quantity and no tuned constant. On EMIT it beats the pool's best heuristic by 4.6% on average (33 of the 40 cells; 8.8% against per-output least squares, and 9.3% on the radiance metric at all ten splits), and the gain is about 5% at every smaller calibration size tried (60, 120 and 240 rows), so it is not bought with the full calibration block. On OCO-2 the same rule loses 1.3% on all thirty cells. The two corpora differ the way the mechanism predicts: EMIT has 285 outputs, heterogeneous members and a large mismatch between the members' training losses and the scored metric, and there the constraint removes estimation variance; OCO-2 has forty outputs, members within a factor of six of one another, and almost no mismatch, and there pooling the weights across outputs is the regularizer that matters. A ladder of seventy cells over the covariance cleaners — the raw sample covariance, the rotationally invariant estimator at several floors, the linear and the analytical nonlinear shrinkage of Ledoit and Wolf [18, 19], and the nonparametric eigenvalue-regularized estimator [20] — shows that inside the nonnegativity constraint every consistent cleaner ties to within a percent at full calibration. What loses is corpus-dependent and smaller than that ladder's spread suggests: on EMIT a floor at one σ^2 and the linear shrinkage each lose about 6% and a floor at $0.2\sigma^2$ about 1%, while on OCO-2 the same cleaners lose at most 1% and the linear shrinkage does not lose at all. This is the mechanism Jagannathan and Ma identified for portfolio weights [21]: a nonnegativity constraint is itself a shrinkage of the covariance, and the nonnegativity constraint rather than the cleaning is what produces the gain. The entrywise bound of Fan, Zhang and Yu [22] applies to exactly the problem (4): for weights on the simplex, the excess of $w^\top \Sigma w$ over its optimum is at most twice the largest entrywise error of the covariance estimate, with neither the member-to-row ratio nor the condition number in it. That bound is about the quadratic objective and not about the mean relative error the tables report, and it does not transfer to it; what it explains is why the constrained problem is stable where the unconstrained one is not. Measured on every output, it holds and is loose by one to three orders of magnitude. The same account explains where the gain comes from: the pairs-bootstrap variance of the least-squares weights at the rows the stack is scored on tracks the cell's gain inside every output group (rank correlation 0.69 across the forty EMIT cells with the group effect removed, and 0.74 on a fresh split). The association is within-split: the same predictor taken from one split and applied to another falls to 0.17, so it describes which cells gain on the rows at hand rather than which cells gain in general. Separately, removing the tenth of rows with the largest gains and losses leaves 87% of the EMIT gain, so the constraint helps broadly across the evaluation rows rather than on a few.

Removing the per-corpus choice. The corpus dependence is removed by two rules that use nothing beyond the calibration rows the weights are already fitted on. They use those rows' labels, as the weights do; what they do not use is any information from the rows the stack is scored on, any held-out block, or any decision made per corpus. A five-fold cross-validation on the calibration rows alone, choosing per cell among the constrained stack, its shrunk form, the sum-to-one rotationally invariant stack and the pool's own heuristic, lands 6.0% above the pool's best on EMIT and level with it on OCO-2, within 0.4% of the oracle of the four. A blend of the constrained and least-squares stacks, with the mixing weight chosen by cross-validated calibration error in the score's own metric over a three-vertex grid that includes the pool's heuristic, keeps the constrained stack where it helps and beats both endpoints where neither does. On a split neither the estimator nor the criterion had seen it reads 5.3% above the pool's heuristic on EMIT and 0.10% above it on OCO-2, with no hand tuning; the earlier squared-loss form of the same criterion reads 4.3% and 0.06% on that split, which is why the criterion is written in the score's own metric. A transductive term using the members' predictions at the scored rows was tried in several forms and did not improve on the rule without it; the rule reported here uses neither a transductive term nor a bootstrap. Below a few hundred calibration rows the blend is unstable and the constrained stack alone is the better choice.

What this changes in the tables. The EMIT stack row of Table 1 is the convex stack fitted per component on the full validation block, which is (4) at its simplest. This section supports that choice of rule — a nonnegative combination rather than an unconstrained one, and no covariance cleaning — but it cannot quantify what a different rule would have done to that row: it uses a larger member pool and fits its weights on a partition of the test block, so its margins are not transferable to Table 1. On OCO-2 the members are homogeneous, the best and worst differing by a factor of six, and every rule in the ladder lands within two percent of every other; the constrained rule is not the one to use there, and this section does not score the per-coordinate selection that Table 3 reports, so it says nothing about it.

7 Ten further emulation configurations

The two problems above stand at opposite ends of one axis: a smooth, six-dimensional, low-rank table where an exact kernel with a learned metric is the strongest single model, and a rougher, twenty-dimensional one where the kernel is useful only inside a learned representation. If that axis orders the comparison, it should order other emulation problems too. Table 5 applies the same family set to ten further configurations, with no tuning per corpus beyond what each family selects on validation.

The corpora are the advection equation with discontinuous initial data from the operator suite of [23], which is also the source of the structural-mechanics problem of [6]; Burgers' equation at three viscosities from PDEBench [24]; three one-step maps from The Well [25] — a two-dimensional turbulent radiative layer, an active-matter system, and a Helmholtz staircase; the ClimSim atmospheric physics emulation task [26]; and the paired correction-coefficient corpus of the physics-guided radiative-transfer work of [4], which is a third radiative-transfer emulation problem and the largest here at 487,790 rows. That last corpus is run in both of its configurations: from the atmospheric state alone, and with the low-fidelity 6S coefficients supplied as additional inputs, which is a different and easier problem.

The configurations are these. Advection is the 200-point input/output pair of the operator suite, used at full resolution with no reduction. Burgers is PDEBench's `1D_Burgers_Sols_Nu`ν release, the map from the initial condition to the solution at $t = 1$ on a grid subsampled by four to 256 cells, the input used raw and the output reduced to 128 principal coefficients. The three Well datasets are one-step maps of the full multi-channel state with 256 principal coefficients in and out: the turbulent radiative layer on its 128×384 grid (density, pressure, velocity), active matter on 256×256 (concentration, velocity, and the two tensor fields), and the Helmholtz staircase on its two pressure channels. ClimSim is the low-resolution release, 124 inputs to 128 tendencies, with 100,000 training rows drawn from the 10.1M available and validation and test taken from the release's own validation and scoring files. The correction-coefficient corpus is the paired 6S/libRadtran Sentinel-2 set, nine state variables and the band wavelength mapping to the three libRadtran coefficients, split by atmospheric state so that all bands of a state travel together; its multi-fidelity configuration adds the 6S coefficients of the same state as inputs.

The families are those of Section 3 with two differences forced by scale. The reference family is a ridge regression linear in whatever inputs the models receive, not the cubic of Table 1: a cubic expansion of a 256-dimensional input is not tractable, and on the two corpora whose inputs are six- and nine-dimensional the linear form is the one that was run. And the kernel solves are exact only when the training block has at most 20,000 rows; above that they use a Nyström approximation on 6,000 uniformly drawn landmarks. That cap binds on ClimSim, on the Helmholtz staircase at 20,384 rows, and on the two correction-coefficient configurations. Those entries are what a capped fit gives; they are not the exact solve, and we do not claim them as a bound on it in either direction, since restricting the fit set both discards information and regularizes. The convex stack and the per-coordinate selection are formed from the kernels, the networks, the residual correction and the feature kernel; the ridge is a reference row and is not one of their members.

What each row averages differs and is stated in the table. Advection, Burgers, ClimSim and the two correction-coefficient configurations have three to five random splits. The Well's releases carry their own train, validation and test files, which are used unchanged, so the three turbulent-radiative-layer runs share one split and differ only in network initialisation; their spread, of order 0.01 points, measures initialisation and not sampling. The active-matter and Helmholtz-staircase rows are single runs.

Three things in the table repeat what the two main problems show. First, the ordering between families follows the structure rather than the corpus. Where the map is smooth and the sample small — Burgers at $\nu = 0.1$ — the exact kernel is at 2.81% against the network's

corpus	n	d	q	runs	ridge (linear)	KRR	KRR, learned metric	network	network ensemble	network + residual KRR	kernel on features	convex stack
Advection	19,000	200	200	5	11.29±0.04	11.36±0.04	11.36±0.04	11.69±0.07	11.50±0.05	11.50±0.05	11.64±0.07	11.36±0.04
Burgers $\nu = 0.1$	7,200	256	128	3	13.89±0.53	2.81±0.28	2.90±0.33	20.34±0.61	19.28±0.09	2.78±0.29	4.92±0.30	2.78±0.29
Burgers $\nu = 0.01$	7,200	256	128	3	33.68±0.28	23.40±0.54	23.58±0.50	36.65±1.95	35.31±0.70	23.12±0.56	29.20±0.59	23.01±0.54
Burgers $\nu = 0.001$	7,200	256	128	3	38.38±0.12	31.66±0.16	33.23±1.73	39.98±0.65	38.89±0.11	31.82±0.11	35.59±0.48	31.34±0.14
Turbulent radiative layer	7,200	256	256	3	31.73	30.31	30.26±0.01	32.17±0.02	31.58±0.02	30.40±0.01	31.08±0.01	30.12
Active matter	14,000	256	256	1	37.24	35.81	35.18	37.31	36.93	35.77	36.22	35.05
Helmholtz staircase	20,384	256	256	1	0.00	0.26	0.07	0.40	0.24	0.23	0.45	0.07
ClimSim	100,000	124	128	5	32.24±0.06	27.20±0.05	26.31±0.04	25.10±0.18	24.43±0.17	24.33±0.13	24.71±0.25	24.25±0.17
RT corrections, state inputs	487,773–487,823	8	3	5	71.54±0.59	62.93±0.51	3.97±0.42	3.75±0.25	3.24±0.11	3.13±0.05	3.52±0.07	3.09±0.05
RT corrections, multi-fidelity inputs	487,773–487,823	11	3	5	33.83±0.26	15.76±1.98	4.60±0.31	2.79±0.14	2.40±0.11	2.34±0.09	2.80±0.09	2.32±0.08

Table 5: Mean relative L^2 test error [%] on ten further emulation configurations, mean $\pm$ standard deviation over the runs available for each. n is the number of examples — a range where the state-level split of the correction-coefficient corpus makes it differ by run — d the input dimension the models receive and q the output dimension they regress. The runs are random splits except for the turbulent radiative layer, whose three runs share The Well's own split and differ in network initialisation, and the two single-run rows. The family set and the validation-based selection are common to every row; the network budget is not, and is 200 epochs at width 512 except on ClimSim (60 at 512) and the two correction-coefficient configurations (120 at 384). The ridge column is linear in the inputs the models receive, unlike the cubic reference of Table 1, and is not a member of the convex stack; its Helmholtz-staircase entry is 0.004, below the two decimals shown. Kernel solves are exact when the training block has at most 20,000 rows and otherwise use 6,000 Nyström landmarks, which is the case on ClimSim, the Helmholtz staircase and the two correction-coefficient rows.

20.34%, a factor of seven, of the same kind as the five- to six-fold gap between the per-input kernel and the network on the EMIT table's smooth components; as the viscosity falls and the solutions roughen the gap closes to 23.40% against 36.65% and then to 31.66% against 39.98%. Where the sample is large and the map rough — ClimSim, and the correction-coefficient corpus at nearly half a million rows — the network leads instead, 25.10% against 27.20% and 3.75% against 62.93%, which is the OCO-2 regime. Where the operator is linear, on advection, the linear ridge is the lowest row and every other family lands within 0.4 points of it, as it should.

Second, a learned metric is what rescues a kernel on the state when the plain one fails, and it is worth most where the plain kernel is worst. On the correction-coefficient corpus from the state alone the isotropic kernel is at 62.93 ± 0.51 and a kernel-flow metric [31] takes it to 3.97 ± 0.42, a factor of sixteen in the mean, which brings a kernel on the state to within six percent of the network on a corpus where the plain kernel was seventeen times worse. With the low-fidelity coefficients supplied as inputs the same pair reads 15.76 and 4.60, a factor of three: the easier the problem is made for the plain kernel, the less the learned metric adds. This is the same mechanism as the per-input length scales on the EMIT table and the empirical-Bayes metric of Section 5.4, at a scale where its effect is unambiguous.

Third, the combination is at or near the best row of its own pool everywhere, and where it is not the lowest row of the table the reason is membership rather than weighting. The convex stack of Section 6 is the lowest row on eight of the ten configurations and the second lowest on the other two; the residual-corrected network is in the lowest two on five. Both exceptions are the linear problems, and on both the row that beats the stack is the ridge, which the stack does not contain. On advection the ridge is lower by 0.07 points while the stack, at 11.36%, matches the best member it has — the isotropic kernel, on which it puts weight 0.87 — to two decimals. On the Helmholtz staircase the ridge reaches 0.004%, sixteen times below the next row of the table and about a hundred times below the weakest member of the pool, and the stack reaches 0.066%, again a shade below its own best member, the learned-metric kernel at 0.067%, on which it puts weight 0.94. In both cases the stack does what it is asked to do inside its pool; what it cannot do is recover a family the pool excludes. The statement this supports is correspondingly narrow: a combination is a good default when no member of its own pool dominates it, and a combination can only be read against the whole comparison when the pool contains the families the comparison includes.

Two limits on scope. These rows use one training budget per corpus and one architecture family, and several of these corpora have published results from methods tuned for them specifically; the comparison here is between families under a common protocol, not against the state of the art on each problem. And with three to five runs, and one on two rows, the spreads support the large orderings above and not fine distinctions between adjacent families.

8 Discussion

The EMIT comparison settles three things, and on each of them the ten-split protocol changes what the single split reported. The first is the standing of the exact kernel. Fitted on a 4,000-point subset with a single length scale it is a useful baseline; fitted on every training row with a length scale per input it is the best single model on the EMIT table, and its metric is interpretable, since the coordinate search finds, from the validation rows alone, that the transmittances and the spherical albedo do not depend on the relative azimuth. The second is the standing of the corrected network. Its mechanism is the one described in [6] — the exact regression of the residuals hands the network the kernel's accuracy on the smooth components while the network keeps its own representation — but with the kernel allowed its own metric the correction is one of four heads within a few hundredths of a percent of one another on the radiance, not the lowest row. The third is the way heads are combined. A per-coordinate selection made on 2,098 validation rows does not transfer to the test block, and its spread across splits is among the largest in the table; a convex stack per component, fitted in the reported metric, is the best row at every split but one. We read this as a statement about calibration data rather than about the heads: with a few thousand rows the right way to use complementary predictors is a low-dimensional combination, not a per-output choice.

The OCO-2 problem says the same thing from the opposite structure. There the kernel on the state is an order of magnitude behind the network, the residual correction adds nothing, and the construction that helps is the exact kernel inside the network's representation. The ridge-readout control settles what that construction is worth and why: the kernel head beats a linear readout refitted on the same frozen features at every split, on every band and under every loss, while the linear readout is worse than the network it reads out from. The gain is the kernel, not the refitting, and it is available to any trained network at the cost of one Cholesky factorization.

The other thing OCO-2 shows is that the training loss, not the architecture, is the axis a selection can exploit. Networks trained on the coefficients and networks trained through the reconstruction sit at opposite ends of the two reported metrics, and because the reconstruction basis is orthogonal a per-coefficient choice acts on both metrics through the same errors, so a choice made on validation can hold both ends at once, as it does here: one coefficient of forty, assigned at every split on every band to the kernel head of the network trained on the weighted coefficients, is what separates a selection that loses the radiance metric from one that wins it. The single-split version of this study reached neither metric and attributed the failure to a member set that varied architecture within one objective; varying the objective instead, as Table 2 does, confirms that attribution.

Section 6 identifies the rule behind that row: the convex stack that wins the EMIT table is the constrained minimum-variance rule in its simplest form; the random-matrix cleaners that seemed the natural refinement act on the covariance of the predictions rather than of the errors; the nonnegativity constraint, and not the cleaning, is what stabilizes the estimate; and on a pool that one head dominates, no combination rule is worth more than a percent. Section 7 applies the same family set to ten further configurations, where the family ordering follows the structure as it does here, a learned metric rescues a kernel on the state exactly where the plain kernel fails worst, and the stack is the lowest row on eight of the ten and the second lowest on the other two, in both cases behind a reference family its pool does not contain.

The spherical albedo is the hard component throughout. Every family in Table 1 is an order of magnitude worse on Y_4 than on the other three, and the per-band anatomy places that error where the albedo is close to zero, so relative error is large there for any predictor. No family in this pool moves it, and larger networks and longer schedules are not evaluated here. An albedo error reaches an emulator used through the inversion (2) directly and everywhere, not only in the cancellation tail: by (3), $\partial\hat{\rho}/\partial\hat{Y}_4 = -\hat{\rho}^2$, so at $\rho = 0.7$ an albedo error moves the recovered reflectance by about half its own size at every band, whatever the transmitted flux. What the tail of Section 2 amplifies is the other three components, whose sensitivities carry the vanishing denominator.

On cost, the exact solve is one Cholesky factorization of an 18,884 $\times$ 18,884 matrix per component and one kernel row per query; the per-input search multiplies the tuning cost by the number of coordinates and leaves the solve unchanged. The networks cost more to train and less to query. The stack costs nothing beyond its members.

Only differences that exceed the seed-to-seed spread should be read as comparisons.

Acknowledgements

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital; both entities supported this work and are named above. These entities and the public funders had no role in the study design, analysis, manuscript preparation, or decision to publish.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author made substantial use of generative AI tools, and describes their role here. The software — the data pipeline, the model comparison, the diagnostics, and the figures — was written and executed by Claude Code (Anthropic) and Codex (OpenAI), which also prepared the first draft of this manuscript, working from the author's data and from the method and experimental recipe of [6]. The measurements of the data's structure, the identification of the ill-posed reflectance entries, and the reevaluation of Section 5.5 were produced in the same AI-assisted sessions and verified against the run summaries. The author reviewed the code, the numbers, and the text, and takes full responsibility for the content of the manuscript.

Data and code availability

The dataset (`data_EMIT_24k`) consists of tabulated radiative-transfer evaluations generated at the Jet Propulsion Laboratory; the OCO-2 data and the reference emulator's stored predictions are those of the companion paper's release; the corpora of Section 7 are public releases, cited there. The pipeline, the per-seed records behind every number reported here (one record per band, seed and family for OCO-2, one per seed for EMIT, and the seventy-cell stacking runs), and the exact configuration of each experiment are retained by the author and will accompany the published version of this manuscript; the method repository of [6] is at `https://github.com/yspennstate/neural-means-kernel-corrections`.

References

[1] A. Berk, P. Conforti, R. Kennett, T. Perkins, F. Hawes, and J. van den Bosch. MODTRAN6: a major upgrade of the MODTRAN radiative transfer code. In *Proc. SPIE 9088, Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XX*, 2014.

[2] D. R. Thompson, V. Natraj, R. O. Green, M. C. Helmlinger, B.-C. Gao, and M. L. Eastwood. Optimal estimation for imaging spectrometer atmospheric correction. *Remote Sensing of Environment*, 216:355–373, 2018.

[3] P. G. Brodrick, D. R. Thompson, J. E. Fahlen, M. L. Eastwood, C. M. Sarture, S. R. Lundeen, W. Olson-Duvall, N. Carmon, and R. O. Green. Generalized radiative transfer emulation for imaging spectroscopy reflectance retrievals. *Remote Sensing of Environment*, 261:112476, 2021.

[4] M. A. A. Mazid and N. Rishe. Multi-fidelity emulation of atmospheric correction coefficients with physics-guided Kolmogorov–Arnold networks. *Remote Sensing*, 18(11):1826, 2026.

[5] O. Lamminpää, J. Susiluoto, J. Hobbs, J. McDuffie, A. Braverman, and H. Owhadi. Forward model emulator for atmospheric radiative transfer using Gaussian processes and cross validation. *Atmospheric Measurement Techniques*, 18:673–694, 2025.

[6] Y. Shmalo. Neural means and kernel corrections for operator learning. Preprint, ResearchGate, 2026. Code and per-run summaries at `https://github.com/yspennstate/neural-means-kernel-corrections`.

[7] L. Berlyand, E. Sandier, Y. Shmalo, and L. Zhang. Enhancing accuracy in deep learning using random matrix theory. *Journal of Machine Learning*, 3(4):347–412, 2024.

[8] Y. Shmalo, J. Jenkins, and O. Krupchytskyi. Deep learning weight pruning with RMT-SVD: increasing accuracy and reducing overfitting. arXiv:2303.08986, 2023.

[9] P. Batlle, M. Darcy, B. Hosseini, and H. Owhadi. Kernel methods are competitive for operator learning. *Journal of Computational Physics*, 496:112549, 2024.

[10] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021.

[11] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[12] G. R. Yoo and H. Owhadi. Deep regularization and direct training of the inner layers of neural networks with kernel flows. *Physica D: Nonlinear Phenomena*, 426:132952, 2021.

[13] R. O. Green et al. The Earth surface mineral dust source investigation: an Earth science imaging spectroscopy mission. In *IEEE Aerospace Conference*, 2020.

[14] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[15] A. Radhakrishnan, D. Beaglehole, P. Pandit, and M. Belkin. Mechanism for feature learning in neural networks and backpropagation-free machine learning models. *Science*, 383(6690):1461–1467, 2024.

[16] V. A. Marchenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.

[17] J. Bun, J.-P. Bouchaud, and M. Potters. Cleaning large correlation matrices: tools from random matrix theory. *Physics Reports*, 666:1–109, 2017.

[18] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[19] O. Ledoit and M. Wolf. Analytical nonlinear shrinkage of large-dimensional covariance matrices. *Annals of Statistics*, 48(5):3043–3065, 2020.

[20] C. Lam. Nonparametric eigenvalue-regularized precision or covariance matrix estimator. *Annals of Statistics*, 44(3):928–953, 2016.

[21] R. Jagannathan and T. Ma. Risk reduction in large portfolios: why imposing the wrong constraints helps. *Journal of Finance*, 58(4):1651–1683, 2003.

[22] J. Fan, J. Zhang, and K. Yu. Vast portfolio selection with gross-exposure constraints. *Journal of the American Statistical Association*, 107(498):592–606, 2012.

[23] M. V. de Hoop, D. Z. Huang, E. Qian, and A. M. Stuart. The cost-accuracy trade-off in operator learning with neural networks. *Journal of Machine Learning*, 1(3):299–341, 2022.

[24] M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert. PDEBench: an extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems 35*, 2022.

[25] R. Ohana, M. McCabe, L. Meyer, R. Morel, F. Agocs, M. Beneitez, M. Berger, B. Burkhart, S. Dalziel, D. Fielding, et al. The Well: a large-scale collection of diverse physics simulations for machine learning. In *Advances in Neural Information Processing Systems 37*, pages 44989–45037, 2024.

[26] S. Yu, W. M. Hannah, L. Peng, J. Lin, M. A. Bhouri, R. Gupta, B. Lütjens, J. C. Will, G. Behrens, et al. ClimSim: a large multi-scale dataset for hybrid physics-ML climate emulation. In *Advances in Neural Information Processing Systems 36*, 2023.

[27] D. H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.

[28] L. Breiman. Stacked regressions. *Machine Learning*, 24:49–64, 1996.

[29] A. G. Wilson, Z. Hu, R. Salakhutdinov, and E. P. Xing. Deep kernel learning. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, pages 370–378, 2016.

[30] J. M. Bates and C. W. J. Granger. The combination of forecasts. *Operational Research Quarterly*, 20(4):451–468, 1969.

[31] H. Owhadi and G. R. Yoo. Kernel flows: from learning kernels from data into the abyss. *Journal of Computational Physics*, 389:22–47, 2019.

[32] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters. Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7):1467–1470, 1999.